

Cyber Attacks Explained

DNS Invasions

We often read about defaced websites whose pages get changed to some malicious content. How do hackers do it and how do we protect our infrastructure from them? This article is about how hackers invade DNS (domain name systems). DNS invasion is technically advanced and a harmful attack on a network or Web infrastructure. Network administrators are urged to learn more about it and strive to secure the infrastructure they manage.



Part—4

As we all know, the DNS exists because it's impossible for humans to remember IP addresses for sites, but easy to remember alphanumeric names. The DNS system was created when the Internet was a friendly place—leading to quite a few issues.

Figure 1 shows how name resolution fundamentally works. When an application (like a browser) wants to connect to a destination service, it queries the DNS server, asking for the IP address. This query is sent over UDP port 53 as a single request and receives a single-packet reply from the server. (Note that since UDP data space is limited to 512 bytes, the protocol stack automatically uses the TCP protocol for queries and replies.) When the client receives a reply, it updates its local cache with the received entry, speeding up subsequent queries to the same domain. Entries in local cache are automatically purged after their TTL (Time to Live) expires.

The DNS uses record types like A, CNAME, SOA, MX, etc. While explaining these is beyond the scope for this article, it is important for administrators to know the usage of each, and before implementing them, they should be evaluated from the security standpoint. Before we learn about DNS-based attacks, we need to know about two types of queries—iterative and recursive.

An iterative DNS query: When a client queries a DNS server, asking if it has the answer for a given domain name, the DNS server may or may not have the answer ready. If the DNS server doesn't have an answer, instead of shutting the request down, it sends the name of an upstream DNS server that might have the answer. This is usually called a DNS referral. The client sends the query to the next (referred) server; if that one too doesn't have an answer, it sends a referral to yet another upstream server. This process continues till either the client gets an IP address or gets a 'query failed' error message.

The recursive DNS query: In this case, the query begins by a client host requesting a name resolution to its immediate DNS server. If the DNS server does not have the answer, it is supposed to do the job of talking to upstream servers, instead of merely providing their referral names. Again, if the upstream server does not have an answer, it needs to take on the responsibility further. This continues till either the root domain server is reached, which must have the answer, or if the queried name itself does not exist, in which case an error message percolates down the chain to the client. Unlike the iterative method, a recursive query proves to be more aggressive in getting query results.